# ENSF 380

## Group Project

## Submission 2 (Week 5)

### Instructions

Work with your assigned group project group. Each team should complete a test worksheet describing proposed tests and submit a PDF file. You should aim to be as comprehensive as possible in describing the tests you would conduct. **You should aim to maximize code coverage and test coverage, describing typical scenarios, edge cases, and boundary conditions.**

In evaluation, we will look for a subset of tests that we expect to be included in a comprehensive set of tests. The specific tests that we will look for will not be named in advance but will be selected from the full set of tests which could have been provided. We will look for the same tests for all groups.

Use the UML diagram we have provided as a solution to group project submission 1 as the basis for completing your worksheet, rather than the one your team created for submission 1. Class relationships depicted in the UML diagram should be considered in this submission (see example below).

Our asking you to use a standard diagram does not imply that your solution was incorrect - there are multiple possible valid solutions to submission 1. Our model solution is one possible design. Furthermore, it is not an optimal design as it has been limited to concepts which have been covered or will be covered shortly.

Standardizing all teams with a single diagram aids us in grading the submission. It also ensures that any changes in the composition of team members will not affect the group project development. Finally, using our model diagram ensures that all teams, regardless of their performance on GP #1, will be working with an acceptable UML diagram for GP #2, thus eliminating cumulative impacts between submissions.

Refer to the lesson Testing Part I (week 4) to complete the work.

The file should be named <Group #>.pdf - for example, group 5 should submit a file 5.pdf.

## Example worksheet entries

This example continues the horse/rider example from submission 1. It shows a *subset* of the unit tests which might be written, given the methods and attributes shown in the UML diagram. In your actual worksheet, you should aim to be as comprehensive as possible.

| **Test Name** | **Description** | **Test Value(s)** |
|---|---|---|
| testRiderSetAge | setAge() should modify the rider's age to the new value. getAge() is used to check the value. | 25 used in constructor, 20 used by setAge() |
| testRiderInvalidAgeBoundary | It should not be possible to instantiate a Rider with an invalid human age (<1 or >120). Check both boundary conditions. | 0<br><br>121 |
| testRiderInvalidAgeNegative | It should not be possible to instantiate a Rider with an invalid human age (<1). Check negative number. | -5 |
| testAnimalViewSkills | Public skills array can be accessed from an instantiated Animal object. | ['dancing', 'karaoke', 'long jump'] |
| testRiderAggregation | A Rider object used in the Animal constructor can be retrieved with getRider(). getRiderName() is used to confirm the Rider object. | Jasmine |

## Support for Wellbeing

Every year, the project in ENSF 380 involves developing software for socially beneficial services. We recognize that some of these topics may be personally challenging or sensitive.

Your well-being is important. If you experience any discomfort or distress while working on your project, please know that support is available. The University of Calgary offers 24/7 mental health support through the Student Wellness Service during working hours and the After Hours Mental Health Support Service.

You can find more information and access these resources here:

https://www.ucalgary.ca/wellness-services/learn/supporting-students-distress